

## /research

Research system: /research create [--mode tiered|supervisor|competitive] ... | list | open | search | show | delete | archive | cluster

### Overview

/research drives the structured research pipeline: web search plus local file cross-referencing, synthesis, and a self-contained RESEARCH.md output. Projects are named, persisted, and can be listed, resumed, clustered, exported, and imported. Create accepts a rich flag set covering depth, tier, mode, per-stage models, concurrency, timeouts, and source limits. The TUI runs the pipeline on a background task while the status bar shows [wait] research: <name>....

### Syntax

```
/research help
/research create <name> <topic> [flags]
/research list [--all] [--json]
/research show <name> [--json]
/research open <name>
/research search <query> [--json]
/research delete <name> [--yes]
/research archive <name>
/research resume <name>
/research continue <name> [--message <text>]
/research update <name>
/research cluster <name> [--force]
/research export <name> [--output <path>]
/research import <path> [--name <name>]
/research config
```

The first positional after create is the project name; everything after it is the topic. The parser splits on whitespace and honours double quotes, so a topic can be quoted as one argument.

### Subcommands

| Form                    | Description                                                 |
|-------------------------|-------------------------------------------------------------|
| create <name> <topic>   | Starts a research project. See the flag table below.        |
| list [--all] [--json]   | Lists projects; --all includes archived, --json emits JSON. |
| show <name> [--json]    | Shows project state and findings; --json emits JSON.        |
| open <name>             | Opens the project's RESEARCH.md in the log panel.           |
| search <query> [--json] | Searches stored research content.                           |
| delete <name> [--yes]   | Deletes a project; --yes skips confirmation.                |
| archive <name>          | Moves a project to the archive.                             |
| resume <name>           | Resumes an interrupted or archived project.                 |

| Form                                                        | Description                                                                                   |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <code>continue &lt;name&gt; [--message &lt;text&gt;]</code> | Continues an existing project with an optional steering message.                              |
| <code>update &lt;name&gt;</code>                            | Refreshes an existing project's findings.                                                     |
| <code>cluster &lt;name&gt; [--force]</code>                 | Runs research clustering on the project; <code>--force</code> re-runs over existing clusters. |
| <code>export &lt;name&gt; [--output &lt;path&gt;]</code>    | Exports the project bundle to a path.                                                         |
| <code>import &lt;path&gt; [--name &lt;name&gt;]</code>      | Imports a project bundle, optionally renaming it.                                             |
| <code>config</code>                                         | Shows the resolved research configuration.                                                    |
| <code>help</code>                                           | Prints the subcommand table.                                                                  |

## Create flags

| Flag                                                 | Values / Default                                                                                                                                        |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--from-url, --from-url (repeatable)</code>     | Seed the run from one or more URLs.                                                                                                                     |
| <code>--from-file, --from-file (repeatable)</code>   | Seed the run from local files.                                                                                                                          |
| <code>--iterations N</code>                          | Iteration budget for the gather loop.                                                                                                                   |
| <code>--depth</code>                                 | <code>shallow</code>   <code>standard</code>   <code>deep</code> .                                                                                      |
| <code>--tier</code>                                  | <code>light</code>   <code>full</code>   <code>dissertation</code> .                                                                                    |
| <code>--mode</code>                                  | <code>tiered</code>   <code>supervisor</code>   <code>competitive</code> .<br><code>competitive</code> implies <code>--format comparison-table</code> . |
| <code>--summarization-model</code>                   | <code>&lt;provider:model&gt;</code> override for summarisation.                                                                                         |
| <code>--research-model</code>                        | <code>&lt;provider:model&gt;</code> override for the researcher stage.                                                                                  |
| <code>--compression-model</code>                     | <code>&lt;provider:model&gt;</code> override for compression.                                                                                           |
| <code>--final-report-model</code>                    | <code>&lt;provider:model&gt;</code> override for the final report.                                                                                      |
| <code>--max-concurrent-research-units N</code>       | Parallel research unit cap.                                                                                                                             |
| <code>--format</code>                                | <code>report</code>   <code>executive-summary</code>   <code>comparison-table</code>   <code>source-bibliography</code>   <code>imrad</code> .          |
| <code>--sources-dir &lt;dir&gt;</code>               | Local sources directory to cross-reference.                                                                                                             |
| <code>--template &lt;name&gt;</code>                 | Output template override.                                                                                                                               |
| <code>--fetch-concurrently N</code>                  | Concurrent fetch cap (default 10).                                                                                                                      |
| <code>--local-concurrently N</code>                  | Concurrent local-file reads (default 8).                                                                                                                |
| <code>--fetch-timeout-secs N</code>                  | Per-fetch timeout (default 30).                                                                                                                         |
| <code>--web-phase-timeout-secs / --web-time N</code> | Web phase budget.                                                                                                                                       |
| <code>--local-phase-timeout-secs N</code>            | Local phase budget.                                                                                                                                     |
| <code>--search-max-retries N</code>                  | Search retry count (default 2).                                                                                                                         |
| <code>--search-retry-base-delay-ms N</code>          | Retry backoff base (default 200).                                                                                                                       |
| <code>--search-circuit-breaker-threshold N</code>    | Consecutive failures before the breaker opens (default 3).                                                                                              |
| <code>--max-web-results N</code>                     | Web result cap.                                                                                                                                         |
| <code>--max-search-calls N</code>                    | Search-call cap.                                                                                                                                        |
| <code>--max-local-sources N</code>                   | Local source cap.                                                                                                                                       |
| <code>--max-synthesis-sources N</code>               | Sources admitted into synthesis.                                                                                                                        |

| Flag                                  | Values / Default                                    |
|---------------------------------------|-----------------------------------------------------|
| <code>--brief &lt;text&gt;</code>     | Provide the brief up front, skipping clarification. |
| <code>--use-local</code>              | Include local file sources.                         |
| <code>--use-specs</code>              | Include spec documents as sources.                  |
| <code>--use-low-relevance</code>      | Keep low-relevance hits.                            |
| <code>--no-papers</code>              | Exclude scholarly-paper engines.                    |
| <code>--use-pdf</code>                | Enable PDF extraction.                              |
| <code>--clarify / --no-clarify</code> | Force on or off the clarification stage.            |
| <code>--evaluate</code>               | Enable evaluation scoring of findings.              |

## Examples

`/research create rust-async "Compare tokio and async-std task scheduling models"`

Names the project `rust-async`, uses everything after it as the topic.

`/research create deep-dive "tree-sitter incremental parsing" --tier dissertation --depth deep`

Runs a deep, dissertation-tier project.

`/research create db-compare "sqlite vs duckdb analytics" --mode competitive`

Competitive mode forces `--format comparison-table` output.

`/research create local-audit "unsafe usage" --sources-dir ./crates --use-local --use-specs`

Cross-references local crates and spec documents.

`/research create fast "tokio broadcast channels" --tier light --fetch-concurrently 20 --web-phase`

A light run with a raised fetch concurrency and web budget.

`/research create rerun "existing project" --summarization-model ollama:qwen2.5:1.5b`

Routes the summarisation stage to a fast local model.

`/research list --all`

Lists every project including archived ones.

`/research continue rust-async --message "Focus on the cancellation paths"`

Continues a project with steering guidance.

## Output

- While running, the status bar shows `[wait] research: <name>...` and the pipeline streams progress lines into the message window.
- On completion the project's `RESEARCH.md` is written and a summary block lists the report path, source counts, and synthesis highlights.
- `list` renders a project table; `--json` variants emit machine-readable output for scripting.
- Failures print the failing stage with a corrective message; the project remains resumable via `/research resume`.

## Related

- `ragent research` CLI runs the same engine outside the TUI.
- `GET/POST/DELETE /research` HTTP endpoints mirror these operations.
- `/research cluster` relies on the research clustering engine.
- See `docs/research-phase2-plan.md` and the research system docs for internals.